



Natural Language Processing

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Part 1: Linear models for Tagging

Tagging tasks in NLP

Log-linear models for Tagging

Tagging Tasks

Tagged Sequences

a b e e a f h j $\Rightarrow$ a/Y b/Z e/Y e/Y a/Z f/X h/Z j/Y

Example 1: Part-of-speech tagging

Profits/N soared/V at/P Boeing/N Co./N ,/, easily/ADV
topping/V forecasts/N on/P Wall/N Street/N ,/, as/P
their/POSS CEO/N Alan/N Mulally/N announced/V first/ADJ
quarter/N results/N ./.

Example 2: Named Entity Recognition

Profits/O soared/O at/O Boeing/B-CO Co./I-CO ,/O easily/O
topping/O forecasts/O on/O Wall/B-LOC Street/I-LOC ,/O as/O
their/O CEO/O Alan/B-PER Mulally/I-PER announced/O first/O
quarter/O results/O ./O

Notation for Tagging Tasks

- ▶ Set of possible input words: $\mathcal{V}$
- ▶ Set of possible tags: $\mathcal{T}$
- ▶ Word sequence: $x_{[1:n]} = [x_1, \dots, x_n]$
- ▶ Tag sequence: $t_{[1:n]} = [t_1, \dots, t_n]$
- ▶ Training data is N tagged sentences, the i^{th} sentence has length n_i :

$$(x_{[1:n]}^{(i)}, t_{[1:n]}^{(i)}) \text{ for } i = 1, \dots, n$$

Independence Assumptions for Tagging

Chain Rule

$$P(t_{[1:n]} \mid x_{[1:n]}) = \prod_{j=1}^n P(t_j \mid t_{j-1}, \dots, t_1, x_{[1:n]}, j)$$

Make independence assumptions

$$P(t_{[1:n]} \mid x_{[1:n]}) \approx \prod_{j=1}^n P(t_j \mid t_{j-1}, x_{[1:n]}, j)$$

j is the word being tagged.

We model the conditional probability directly: no Bayes Rule here.

Questions

- ▶ Split up $P(t_j \mid t_{j-1}, x_{[1:n]}, j)$ into parameters?
- ▶ How to find $\arg \max_{t_{[1:n]}} P(t_{[1:n]} \mid x_{[1:n]})$?

Tagging tasks in NLP

Log-linear models for Tagging

Representation: finding the right parameters

Problem: Predict ?? using context, $P(?? \mid \text{context})$

Profits/**N** soared/**V** at/**P** Boeing/**??** Co. , easily topping forecasts on Wall Street , as their CEO Alan Mulally announced first quarter results .

Representation: history

- ▶ A history is a 3-tuple: $(t_{-1}, x_{[1:n]}, i)$
- ▶ t_{-1} is the previous tag (we are assuming a bigram model)
- ▶ $x_{[1:n]}$ are the n words in the input
- ▶ i is the index of the word being tagged
- ▶ For example, for $x_4 = \text{Boeing}$:
 - ▶ $t_{-1} = \text{P}$
 - ▶ $x_{[1:n]} = (\text{Profits}, \text{soared}, \dots, \text{results}, .)$
 - ▶ $i = 4$

Feature-vectors over history-tag pairs

Take a history, tag pair (h, t)

$f_k(h, t)$ for $k = 1, \dots, m$ are **feature functions** representing the tagging decision.

Example: Part-of-speech tagging [Ratnaparkhi 1996]

$$f_{100}(h, t) = \begin{cases} 1 & \text{if current word } x_i \text{ is Boeing and } t = N \\ 0 & \text{otherwise} \end{cases}$$

$$f_{101}(h, t) = \begin{cases} 1 & \text{if } t_{-1} \text{ is P and } t = N \\ 0 & \text{otherwise} \end{cases}$$

Log linear model for Tagging

- ▶ Let there be m features, $f_k(\mathbf{x}, \mathbf{y})$ for $k = 1, \dots, m$
- ▶ $\mathbf{x} = x_{[1:n]}$ and $\mathbf{y} = t_{[1:n]}$
- ▶ Define a parameter vector $\mathbf{w} \in \mathbb{R}^m$
- ▶ Each $(\mathbf{x}, \mathbf{y})$ pair is mapped to score:

$$s(\mathbf{x}, \mathbf{y}) = \sum_k w_k \cdot f_k(\mathbf{x}, \mathbf{y})$$

- ▶ Using inner product notation:

$$\begin{aligned}\mathbf{w} \cdot \mathbf{f}(\mathbf{x}, \mathbf{y}) &= \sum_k w_k \cdot f_k(\mathbf{x}, \mathbf{y}) \\ s(\mathbf{x}, \mathbf{y}) &= \mathbf{w} \cdot \mathbf{f}(\mathbf{x}, \mathbf{y})\end{aligned}$$

- ▶ To get a probability from the score: Renormalize!

$$\Pr(\mathbf{y} \mid \mathbf{x}, \mathbf{w}) = \frac{\exp(s(\mathbf{x}, \mathbf{y}))}{\sum_{\mathbf{y}'} \exp(s(\mathbf{x}, \mathbf{y}'))}$$

Feature functions for Tagging

Problem

- ▶ We have defined a log-linear model using feature functions:
 $\mathbf{f}(\mathbf{x}, \mathbf{y})$
- ▶ We have defined parameters using a history h so feature functions are: $\mathbf{f}(h, t)$

Locally normalized log-linear taggers

Conditional Distribution over history, tag pair (h, t)

$$\log \Pr(t \mid h) = \mathbf{w} \cdot \mathbf{f}(h, t) - \log \sum_{t'} \exp(\mathbf{w} \cdot \mathbf{f}(h, t'))$$

- ▶ $\mathbf{f}(h, t)$ is a vector of feature functions
- ▶ $\mathbf{w}$ is the weight vector

Local normalization for tagging

- ▶ Word sequence: $x_{[1:n]}$ and tag sequence: $t_{[1:n]}$
- ▶ Histories $h_i = (t_{i-1}, x_{[1:n]}, i)$

$$\log \Pr(t_{[1:n]} \mid x_{[1:n]}) = \sum_{i=1}^n \log \Pr(t_i \mid h_i)$$

Globally normalized log-linear taggers

Global feature function $\Phi(\mathbf{x}, \mathbf{y})$

- ▶ Word sequence: $\mathbf{x} = x_{[1:n]}$ and tag sequence: $\mathbf{y} = t_{[1:n]}$
- ▶ From *local* histories $h_i = (t_{i-1}, x_{[1:n]}, i)$ to global Φ values:

$$\Phi_k(x_{[1:n]}, t_{[1:n]}) = \sum_{i=1}^n f_k(h_i, t_i)$$

- ▶ $\Phi(\mathbf{x}, \mathbf{y}) = (\Phi_1, \Phi_2, \dots, \Phi_m)$ is a *global* feature vector
- ▶ $\mathbf{w}$ is the weight vector for Φ

Global normalization for tagging

$$\log \Pr(\mathbf{y} \mid \mathbf{x}, \mathbf{w}) = \mathbf{w} \cdot \Phi(\mathbf{x}, \mathbf{y}) - \log \sum_{\mathbf{y}'} \exp(\mathbf{w} \cdot \Phi(\mathbf{x}, \mathbf{y}'))$$

Conditional Random Field

Global normalization for tagging

$$\log \Pr(\mathbf{y} \mid \mathbf{x}, \mathbf{w}) = \mathbf{w} \cdot \Phi(\mathbf{x}, \mathbf{y}) - \log \sum_{\mathbf{y}'} \exp(\mathbf{w} \cdot \Phi(\mathbf{x}, \mathbf{y}'))$$

- ▶ This model is also called a conditional random field (CRF)

Algorithms for training and decoding

- ▶ Global normalization could be expensive: requires sum over exponentially many terms $\mathbf{y}'$
- ▶ Finding $\arg \max_{\mathbf{y}} \log \Pr(\mathbf{y} \mid \mathbf{x})$ can be accomplished using the Viterbi algorithm.
- ▶ Training: finding the weight vector $\mathbf{w}$ can be done using a variant of the Forward algorithm.

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